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Water Ice and the Shallow Regolith Structure of the Shackleton Crater, the Moon: Implications for Future Chang'E-7 In Situ Radar Observation

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The unique lighting conditions and environmental characteristics of the lunar polar region create favorable conditions for the accumulation and preservation of water ice, making detecting water ice in the polar permanent shadow regions (PSRs) of significant scientific and practical importance. The Shackleton crater, located near the south pole of the Moon, is believed to be rich in large amounts of water ice deposits within its PSRs, making it a promising potential landing site for the future Chinese Chang'E-7 (CE-7) mission. This paper provides a comprehensive review of water ice detection methods in the Shackleton crater, encompassing Moon-based radars, neutron detectors, spectrometers, etc. The results suggest the probable presence of water ice in the Shackleton crater, particularly within its PSRs, possibly distributed locally from the surface to several meters deep. However, current remote sensing techniques have limitations, necessitating future in situ exploration missions to directly confirm the presence of water ice. This paper aims to contribute to the detection of water ice and the assessment of the regolith structure using the CE-7 ground-penetrating radar, providing a comprehensive reference for future investigations in the Shackleton crater.

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Introduction

The environmental characteristics of the permanent shadow region (PSR) in the lunar south pole create favorable conditions for the accumulation and preservation of water ice, making the detection of water ice in these areas of the utmost engineering value and scientific significance [1,2]. Specifically, it can provide water resources for the future human establishment of lunar bases. The fourth phase of the Chinese Chang'E project includes the establishment of an international lunar base in about 2030. Among the missions of the fourth phase of lunar exploration, Chang'E-7 (CE-7) is expected to be launched around 2026, and its main task is to plan a comprehensive exploration at the south pole of the Moon [3], which includes the topography of the Moon, material composition, space environment, and in situ detection of the presence of water ice in the lunar polar region [3–5]. Water ice resources play a prominent and critical role as an important material and energy fuel (e.g., propellant, oxygen, drinking water, etc.) for deep space exploration activities, especially for manned exploration activities [1,6] and to reduce the cost and risk of human exploration of the Moon and other celestial bodies. Water ice resources are undoubtedly the most important in situ resources on the Moon [2,7]. The Shackleton crater is recommended as a potential landing area for future Chinese CE-7 missions because of the potential for significant water ice in the crater [5,8].

As one of the lunar craters of great scientific significance, the Shackleton crater is located in the south pole–Aitken basin of the Moon, with longitude and latitude of 89.655 °S, 129.174 ° E, almost coinciding with the axis of rotation of the Moon, with a diameter of 21 km and a depth of ~4.1 km [9]. The geological age of the Shackleton crater is constrained to be about 3.6 billion years, and its floor has remained in shadow regions for at least 2 billion years [10,11]. As such, the Shackleton crater may have been accumulating volatiles over these 2 billion years, possibly from solar wind injections [12], and potentially late volcanic activity [13,14]. The south pole of the Moon has a complex topography and large undulations (Fig. 1A), with a maximum elevation of ~1,958 m and mean depth from rim to floor value of about 4.1 km in and around the Shackleton crater (Fig. 1B)

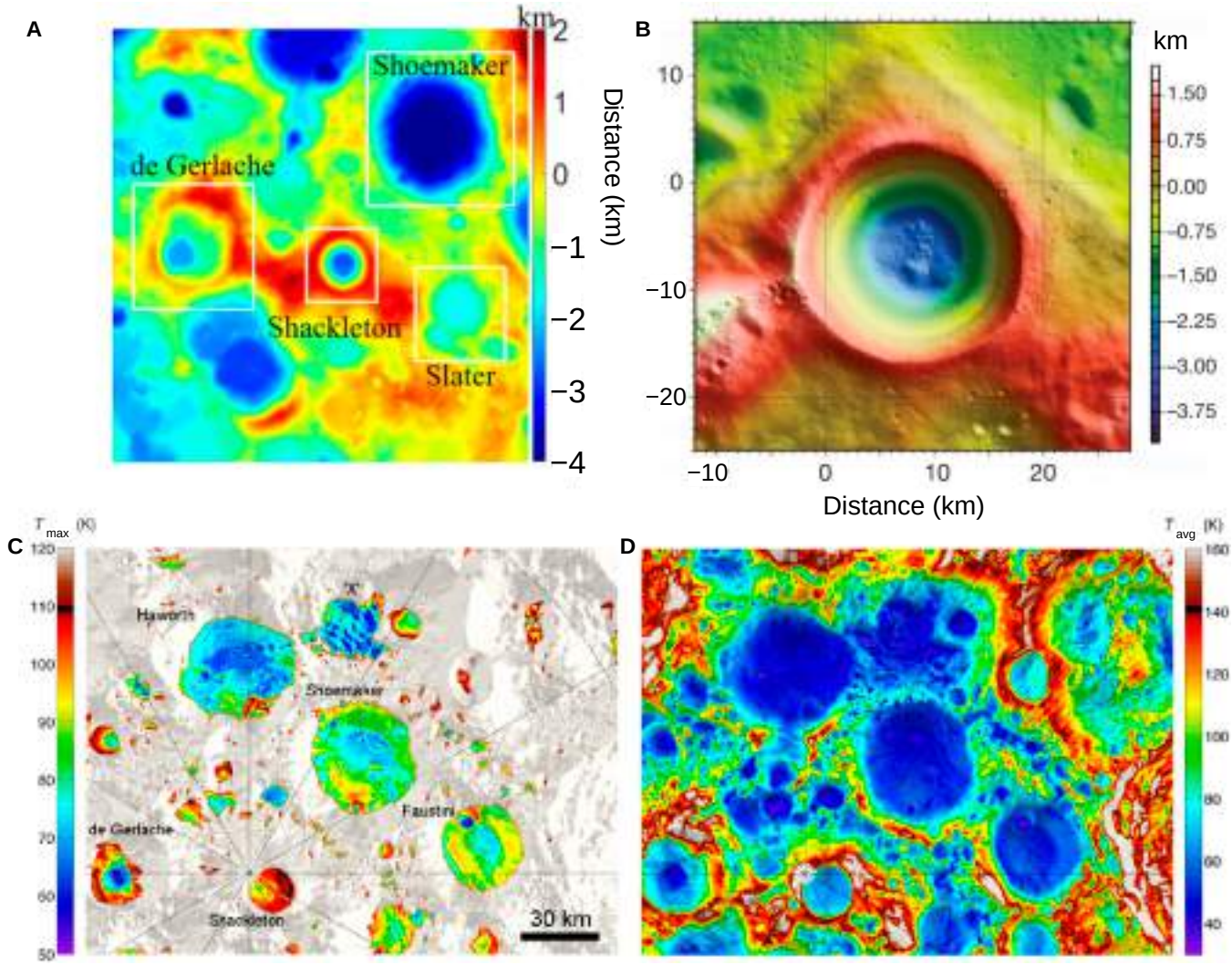

Fig. 1. Digital elevation model (DEM) and temperature distribution of the Shackleton crater of the lunar south pole. (A) DEM of the southern polar region of the Moon [16]. (B) DEM of the Shackleton crater [9]. (C) Annual maximum temperature in the southern polar region of the Moon [15]. (D) Annual average temperature [15].

[9]. The maximum temperature of the Shackleton crater is about 95 to 105 K, with an average temperature throughout the year of about 70 K (see Fig. 1C and D) [15].

Using the data obtained by the Lunar Orbit Laser Altimeter (LOLA) mounted on the Lunar Reconnaissance Orbiter (LRO), it has been demonstrated that the Shackleton crater displays its maximum slope at the crater wall, which can reach up to approximately 35° [9]. The bottom of the crater is the flattest, where the maximum slope is about 10°. The slope of the crater's edge falls between the slope of the crater wall and the slope of the crater floor, with a maximum slope of about 20° [9,16]. The Selenological and Engineering Explorer (SELENE) spacecraft utilized the reflection of the crater wall to image the shadow region within the crater, revealing a limited number of small craters with diameters no greater than a few hundred meters [17]. The bottom of the crater is covered with an undulating hill-like structure approximately 300 to 400 m in thickness, with a central peak that attains a height of approximately 200 m [17]. Previous lunar observations by space-borne synthetic-aperture radar (SAR) have shown that the PSR of the Shackleton crater has a high circular polarization ratio (CPR) value. This high CPR value is suggestive of water ice resources, which are postulated to exist within the PSR surface region of the lunar regolith [18]. However, a high CPR value is not necessarily evidence of the presence of water. It may also be due to the presence of coarse material at the bottom of the crater [19,20]. Detecting polar water ice has been a primary research objective of lunar exploration (e.g., the future CE-7 mission), and the identification of water ice resources on the Moon could provide vital resources for life support during astronaut scientific activities and long-term exploration in the future Moon base. Nevertheless, there is currently inadequate in situ evidence to support the claim that water ice exists within the PSR of the Shackleton crater. Therefore, conducting in-site exploration in the future is highly necessary.

This paper comprehensively reviews the research progress regarding the detection of water ice on the south pole of the Moon, particularly in the PSR of the Shackleton crater, using various detection methods, encompassing Moon-based radars, neutron detectors, spectrometers, etc. [18,21–24]. Given the potential for water ice deposition in the lunar regolith of Shackleton's PSR, radar detection is deemed particularly promising due to its unique penetrating properties. To achieve this objective, we propose that the upcoming CE-7 mission

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should carry a ground-penetrating radar (GPR) with multiple-frequency channels for in situ detection and investigation of the PSR of the Shackleton crater to determine the presence and distribution of water ice. The content of this paper will serve as a preliminary reference for this investigation. The abundance of water ice in the Moon's polar region provides a valuable opportunity to study the Moon's internal materials' composition, the source of water, and the isotopic composition of hydrogen and oxygen in the early Solar System, thereby offering significant scientific insight [5].

Evidence of Water Ice in the Shackleton Crater

Previous studies have carried out a series of water ice detection in the south pole region of the Moon, including radar, neutrons, near-infrared spectroscopy, and far-ultraviolet detection. The results of polar water ice have been reported in relevant papers, and this section introduces the detection of polar water ice and its current research status from the classification of detection methods [18,21–25].

Methods for detecting water ice

Radar observation

Water ice exhibits the property of total internal reflection, preserving the circular polarization direction of transmission in the scattered signal. Additionally, the transmission loss of frozen volatiles is much lower than that of basalt rocks, resulting in a higher average reflectivity of electromagnetic waves and echo energy. As a consequence, water ice can reflect more energy of electromagnetic waves than surface rocks [26]. Surface rocks scatter radar electromagnetic waves in all directions, and some of the energy may not be received by receiving antennas. Therefore, lunar regolith and water ice can be distinguished based on the different polarization modes and energy characteristics of the echoes returned from the 2 surfaces [27]. Water ice deposits are known to have high CPR value [19,28], which results from the volume backscattering effect and the coherent backscattering opposition effect [29,30]. During the detection of polar water ice by spaceborne radars on the Moon, such as Mini-SAR and Mini-RF radars, the CPR value of the radar echo serves as an important parameter indicating the presence of water ice [18,31]. The original radar antenna emits circularly polarized electromagnetic waves (e.g., left-handed), receives a pair of orthogonal linear electromagnetic echoes, and uses the received electromagnetic echo polarization radiation energy to calculate the CPR value, $CPR = \frac{SC}{OC}$, where SC is the total energy received in the same direction and OC is the total energy received by opposite polarization [27,31,32].

Clementine's radar observation of the south pole of the Moon indicated that the Shackleton crater had a high CPR value, which scientists interpreted as the presence of water ice in the crater [26,31]. However, other scientists have raised alternative hypotheses, suggesting that the high CPR values may also result from factors, e.g., surface roughness or crater slopes, though the presence of water ice is not excluded [25,33,34]. Spudis et al. [18] further analyzed Mini-RF data and identified multiple craters in the south pole of the Moon with abnormal CPR values, where the ratio of external CPR to internal CPR was greater than 0.4 (Fig. 2A). These correspondences are consistent with the explanation that high CPR material is water ice, and these 3 variables show plausible correlations, further supporting the idea that anomalous polar craters may contain water ice. Spudis et al. [18] also found that all 24 anomalous craters on the south pole of the Moon are in PSRs, 14 of which were consistent with simulated water ice positions observed by the Lunar Prospector neutron spectrometer (Fig. 2B) [18,35]. The causes of elevated CPR are not singular. Three different surface conditions could lead to high CPR on the Moon, e.g., rough, icy, and multibounce [18,21,36]. By simultaneously using 4 variables (SC enhancement, OC enhancement, CPR, and total power enhancement), they can better differentiate roughness and ice content compared to using CPR alone [18]. Based on this hypothesis, they classified the Shackleton crater as a multibounce condition and suggested that Shackleton's high CPR was likely caused by a combination of surface roughness and water ice [36].

Detection of neutrons and hydrogen

Neutrons are released when high-energy cosmic rays interact with the lunar surface material. Some neutrons with high energy that escapes directly from the lunar surface into the universe are called fast neutrons (500 keV to 8 MeV) [37]. Some neutrons penetrate lunar surface materials and collide with other atoms. If they do not lose much energy during the collision process, they retain a certain amount of energy when they reach neutron detectors, and they are referred to as epithermal neutrons (0.3 eV to 500 keV). If a specific region on the lunar surface has a high concentration of hydrogen, any neutrons moving in this area will undergo rapid "cooling" before escaping from the lunar surface into the universe, meaning their energy will sharply decrease, transforming them into thermal neutrons (<0.3 eV) [37,38]. Therefore, a decrease in fast and epithermal neutrons and a relative increase in thermal neutrons can indicate an increase in hydrogen content, suggesting the presence of water ice on the lunar surface [39,40]. Miller et al. [41] used statistical likelihood analysis on the neutron datasets obtained by the LRO mission to describe the abundance distribution of southern polar hydrogen of the Moon [41]. The hydrogen distribution can be converted into the water content (e.g., Fig. 3) [22,41]. In general, the statistical significance (e.g., the λ-statistic) is proportional to the reduction in neutron count rate (fast neutrons and epithermal neutrons) and the reduction in neutron count rate is proportional to the hydrogen abundance [37]. As shown in Fig. 3C, the relatively high value of the λ-statistic, i.e., the local reduction of fast neutrons, indicates a significant hydrogen enhancement within tens of centimeters of the surface in the Shackleton crater, and the result shows the surface concentration of 0.72 ± 0.13 wt% of water-equivalent hydrogen in the PSR within the Shackleton crater [41].

Near-infrared spectroscopy and far-ultraviolet detection

Infrared spectroscopy detection is essentially a method that determines the molecular structure of substances, identifies compounds, and performs quantitative analysis based on the relative vibrations between atoms within molecules and molecular rotations. One advantage of near-infrared spectroscopy is that OH and other forms of H_2O (e.g., liquid, surface adsorption, or bound in minerals) can be distinguished by directly measuring molecular vibration [42–44]. Near-infrared spectroscopy data from the Moon Mineralogy Mapper (M3) on the Chandrayaan-1 spacecraft provide the highest spatial and spectral resolution near-infrared data, including the polar regions. However, the wavelength range of M3 (0.46 to 2.98 μm) is too

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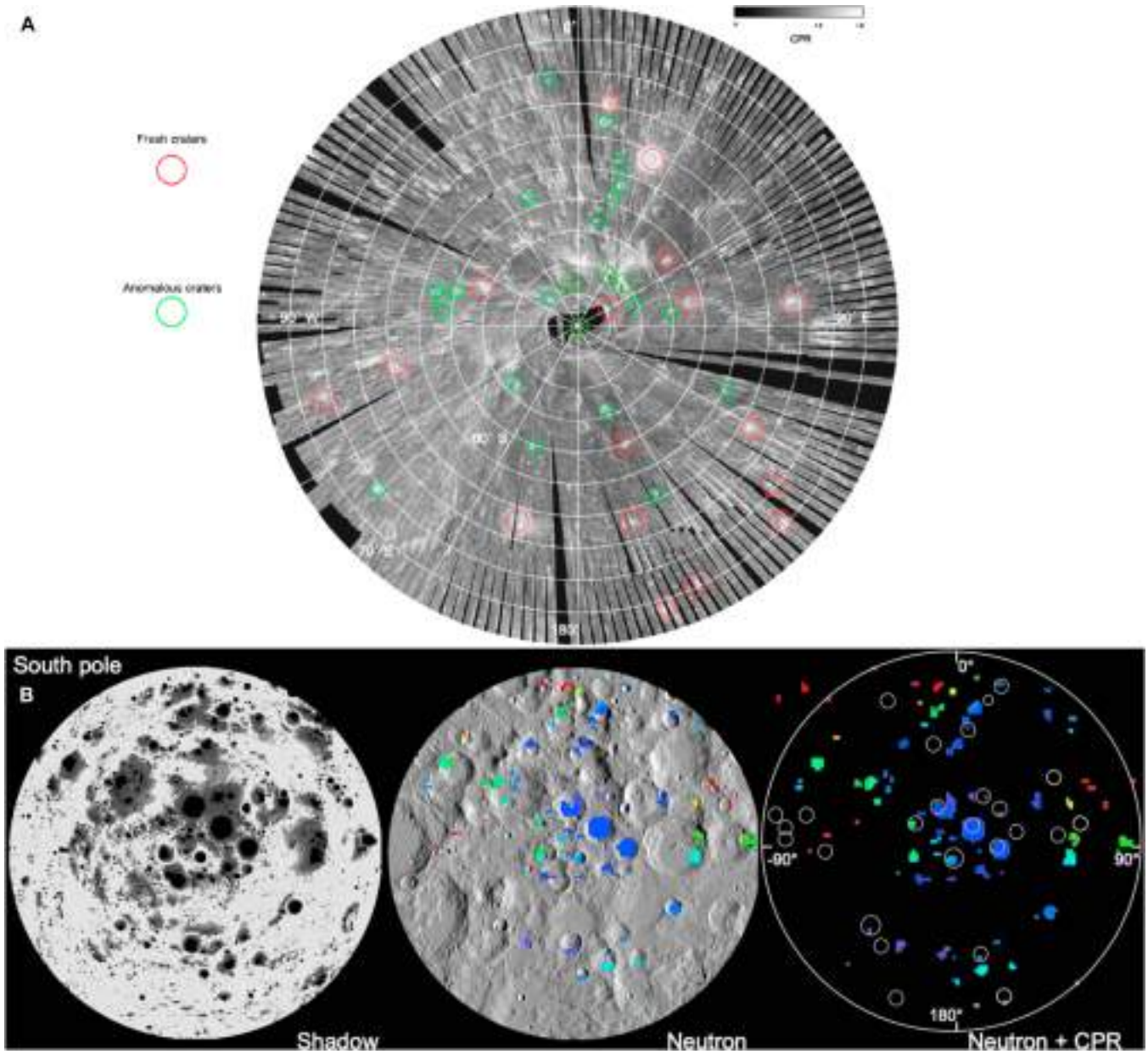

Fig. 2. Map of potential water ice distribution area on the southern pole of the Moon [18]. (A) Map of the distribution of anomalous craters, with green representing anomalous craters. (B) On the left is a map of the distribution of PSR on the southern pole of the Moon (black areas), in the middle is a Pixon model map of the neutron spectral data (purple and blue for small amounts of water-equivalent hydrogen and red for large amounts of water-equivalent hydrogen) [35]. On the right is a composite of neutrons and anomalous CPR craters.

limited to correctly distinguish OH and H_2O using basic vibration modes in the 3-μm wavelength [38,43,45]. Therefore, an improved thermal model suggests that the simultaneous occurrence of absorption in the 3 wavelength ranges around 1.3, 1.5, and 2.0 μm would be evidence of the presence of water ice [23]. Most of the water ice positions observed by the M3 are broadly consistent with the reflectivity values of the LOLA and Diviner's data, as well as the ultraviolet ratio of the Lyman Alpha Mapping Project (LAMP) instrument, providing evidence of the presence of water ice [23]. Further from the perspective of near-infrared spectroscopy (Fig. 4A) is the possible presence of water ice in the Shackleton crater [23], a potential landing area of the CE-7 mission.

The LAMP instrument is a far-ultraviolet imaging spectrometer onboard LRO [46]. The LAMP uses starlight and interplanetary hydrogen emission as illumination sources to measure the ultraviolet reflectance of the lunar night surface and PSR [15,47]. During the LRO mission, the LAMP instrument observed far-ultraviolet emission from the far side of the Moon, and its objective was to search for signs of water frost and other surface volatiles in the PSR. As shown in Fig. 5, off-band (155 to 190 nm) albedo observed by LAMP is consistent with the flat on-band (130 to 155 nm) albedo consistently indicating the presence of ~2.0% water frost in the Shackleton crater [48]. Meanwhile, because most of the Shackleton region had high ultraviolet reflectance, suggesting that this anomalous spectrum was inconsistent

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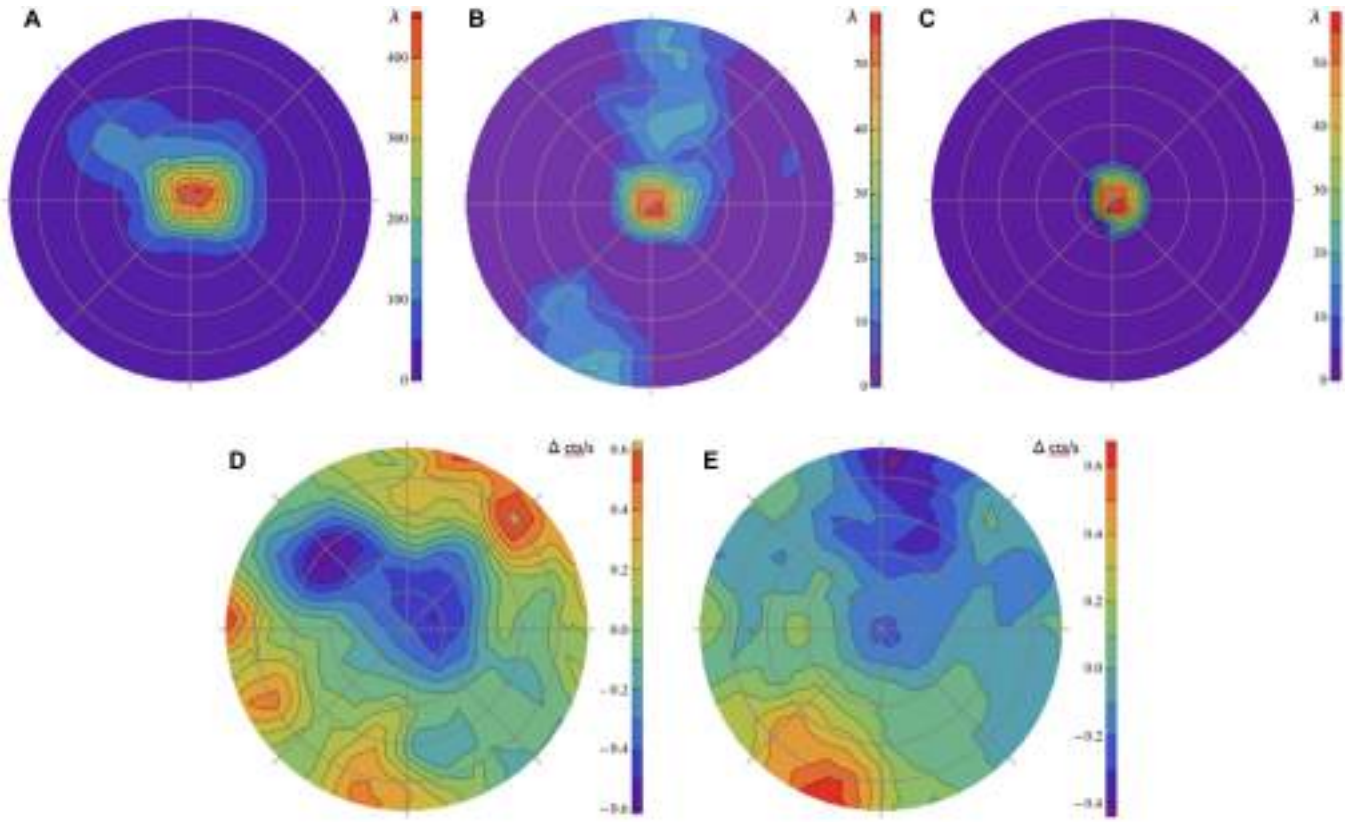

Fig. 3. Spatial distribution of λ-statistics and observed count rate differences on the south pole of the Moon [41]. (A) Distribution of λ-statistics for superheated neutrons. (B) Distribution of λ-statistics for fast neutrons. (C) Fast neutrons decrease region. (D) Count rate difference for superheated neutrons relative to the average value. (E) Count rate difference for fast neutrons relative to the average value.

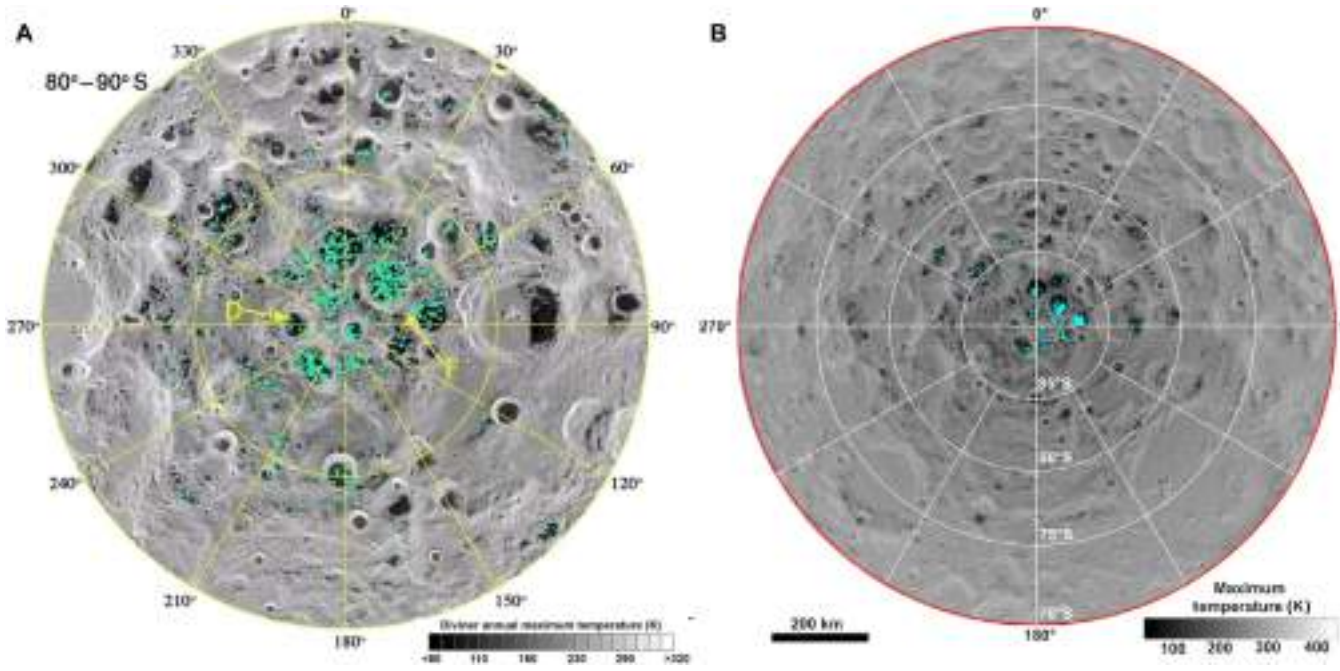

Fig. 4. (A) Distribution of water ice (green dots) in the southern polar region [23]. (B) Detection map of anomalously bright pixels in the southern polar region [24]. Cyan pixels indicate locations with maximum temperatures less than 110 K and slopes less than 10°, and these locations are likely to contain water ice.

with the spectrum of water ice, they could not rule out the mixing of water ice with plagioclase feldspar in the Shackleton crater [15,49].

Temperature and reflectance

In addition to the aforementioned methods, temperature and reflectance are also indicators for detecting the presence of

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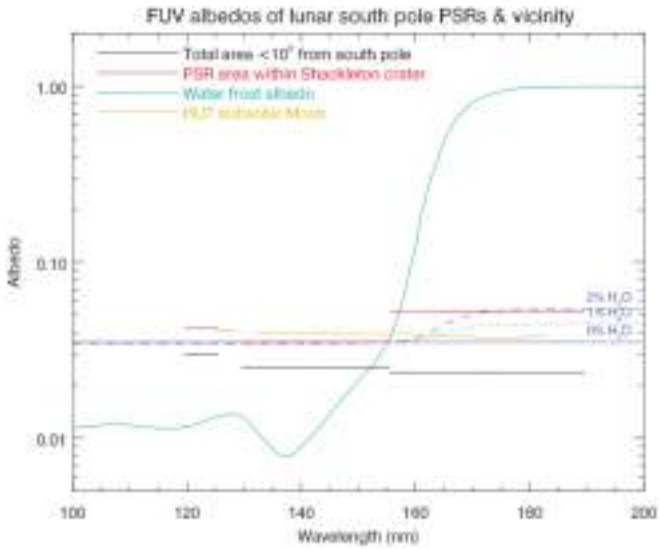

Fig. 5. Nighttime FUV albedo versus wavelength for the PSR of the Shackleton crater [48], also compared with HUT subsolar reflectance [83].

water ice in polar. The concept of cold traps was introduced by Watson et al. [6] and Vasavada et al. [50], suggesting that a sufficiently cold thermal environment on the lunar south pole could allow for the long-term stability of water ice. The sublimation rate of water ice is exponential concerning temperature and is such that surface water ice cannot be preserved on surfaces at temperatures well above 100 K due to rapid sublimation losses [51]. In contrast, surfaces that never exceed temperatures ~100 K can preserve surface frost to prevent sublimation for geologically significant periods. For example, a 1-mm-thick layer of water ice can last more than 2 billion years at 100 K, whereas the same layer can only last about 1,000 years at 125 K. Even a short time at high temperatures will completely sublimate surface deposits, depending on their thickness [52,53]. Fisher et al. [24] utilized reflectivity data from the LOLA and the Diviner's radiometer experiment to observe that the reflectance of 1,064 nm from LOLA suddenly increases when the surface temperature, as measured by Diviner, is below approximately 110 K in the vicinity of the lunar south pole (above 85°). This phenomenon is consistent with the presence of persistent surface water ice deposits. This argument reflects the views of Paige et al. [54], who believe that the sudden increase in reflectance at 110 K is related to the presence of water ice on Mercury. Based on this reasoning, Fisher et al. [24] have created a map depicting the partial distribution of suspected ice-rich locations in lunar polar regions (Fig. 4B). Although the reflectivity of the Shackleton crater may be influenced by bright geological features, e.g., fresh craters, crater rays, ejecta, and plagioclase [49], they do not exclude the possibility of the presence of water ice in Shackleton crater [9,55].

Distribution characteristics and content of water ice

Water ice within the PSR of the lunar south polar region is predominantly present as dirty ice, which probably is a mixture of ice and lunar regolith. Large exposed deposits of pure water ice at the floor of the craters are absent, and the area of exposed surface ice is likely less than a few percent [17]. Previous studies indicate that the weight of water ice present in the crater floor as dirty ice ranges from 1% to 2%, and a thin layer of lunar regolith may cover the top of the dirty ice [17,21]. Although different measurements suggest that H, OH, or water ice is present in the lunar polar regions, determining its location, form, and abundance in specific reservoirs has proven elusive. Mini-RF radar observations indicate that the CPR on the walls of the Shackleton crater is patchy and unevenly enhanced, and its intensity decreases with depth at the floor of the crater [36]. It suggests that the properties of the CPR are likely consistent with the immature roughness effect caused by weathered rock layers. Thomson et al. [36] analyzed the water ice content of the Shackleton crater based on Mini-RF orbital radar observations, as shown in Fig. 6. A comparative analysis of CPR values yielded a maximum water ice content of ~5 to 10 wt% at Shackleton and that water ice deposits are likely distributed in the upper 1 to 2 m of the lunar regolith at the Shackleton crater [36]. This result is consistent with the ~0.7 wt% water equivalent hydrogen found within tens of centimeters of the surface of the Shackleton crater by Miller et al. [41]. In addition, the above result is also consistent with that measured by the LAMP, which shows the presence of ~ 1% to 2% water frost on the surface of the SPR [48].

Shallow Subsurface Structure

As one of the potential landing areas for the CE-7 mission, the internal structure of the lunar regolith and its subsurface structure for the Shackleton crater is an important scientific target for exploration. Halim et al. [56] suggested that most of the impact melt during the formation of the Shackleton crater remained inside the crater. The crater walls are surrounded by concentric fractures, and the material distributed on the crater walls may be exposed to the lunar crustal material. The layered structure material is evident in the crater walls as observed by image data (Fig. 7A) [57]. The ejecta around the Shackleton crater is estimated to be about 150 m thick by the impact crater ejecta model [56]. Also, some landslide material is present at the crater rim of the Shackleton crater [9]. However, the detection of material below the lunar surface of the Shackleton crater is currently almost a gap, especially the data from in situ detection. For the shallow subsurface structure, the main methods are drilling sampling and in situ radar penetration [58,59]. Currently, no drilling sampling data are available for the Shackleton crater. For radar detection, only satellite-based radars (e.g., Mini-RF and Mini-SAR) have ever detected the region, but these 2 radars are SAR radars for surface imaging mapping. However, the ALSE radar carried by the Apollo 17 mission and the LRS radar carried by the SELENE mission performed a global sounding of the Moon but the vertical resolution of these 2 radars was too large to detect the shallow subsurface within tens of meters. Therefore, no radar equipment has been currently used to detect the shallow subsurface structure of the Shackleton crater. This leads to the shallow structure of the Shackleton crater remaining a mystery, waiting to be studied.

To investigate the presence and form of water ice within the Shackleton crater, previous studies have modeled the shallow subsurface structure of the lunar regolith with water ice shown in (Fig. 7B) [21]. This model suggests that the water ice present inside the PSR may be several meters thick, covered with a thin regolith layer (about a few centimeters) above it. This model differs in regolith structure observed by the Moon-based GPR of Chang'E-3, -4, and -5, which were deployed on the near side and far side of the Moon [58,60–63]. The reason is that there is no water ice material in the shallow surface layer of Chang'E-3,

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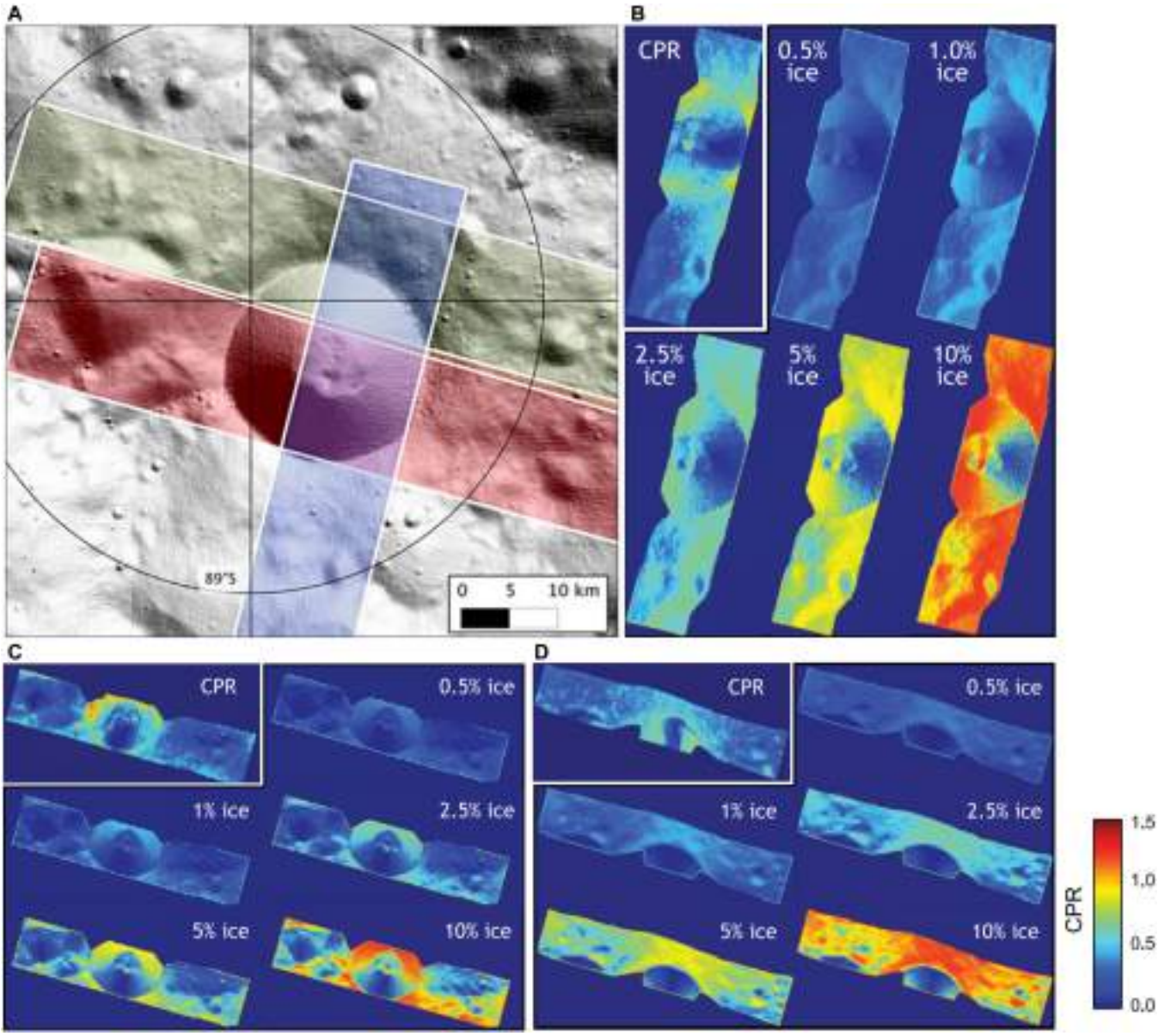

Fig. 6. (A) Mini-RF radar observations of the Shackleton crater [36]. The red color represents the location of the 2010 April 18 observation. The yellow color represents the location of the 2010 June 25 observation, and the blue color represents the 2009 December 23 observation. (B) Radar image of the Shackleton observed on 2010 April 18; 0.5%, 1.0%, 2.5%, 5%, and 10% water ice content are given. (C) Observations on 2009 December 23. (D) Observations on 2010 June 25.

-4, and -5 landing areas. Whether the model of the shallow surface layer of water ice in the polar region is applicable or correct is yet to be determined by the CE-7 mission with the in situ GPR to detect the shallow subsurface structure of the Shackleton crater.

The Future CE-7 Mission

On the afternoon of 2022 November 24, during the UN/China Workshop on Space Exploration and Global Partnerships, the China National Space Administration released an announcement regarding the opportunity for payloads on the CE-7 mission [4]. The CE-7 mission is scheduled to launch around 2026. It aims to explore the environment and resources in the lunar south pole region. This includes tasks such as investigating the lunar south pole surface environment, lunar regolith water ice, volatile components, and more. The mission aims to obtain remote sensing and in situ scientific data from the entire Moon, landing areas, and reconnaissance areas. It also aims to lay the foundation for the construction of an international Moon base [4]. The landing site of the CE-7 mission is selected within the South Pole-Aitken Basin above 85°S of the Moon (Fig. 8C), where the Shackleton crater is considered as one of the potential landing areas for its excellent light conditions (Fig. 8A) and scientific value. In particular, the crater rim of the Shackleton crater can be illuminated by the Sun almost the whole year round (Fig. 8B), providing a good recharge of solar energy for the exploration payload [5].

The PSR of the Shackleton crater is mostly located on the floor and is characterized by no light, ultralow temperature, and no direct communication to Earth due to the occlusion of the crater wall. China's Chang'E project plans to use the CE-7 rover and/or leaper to conduct in situ exploration of the PSR

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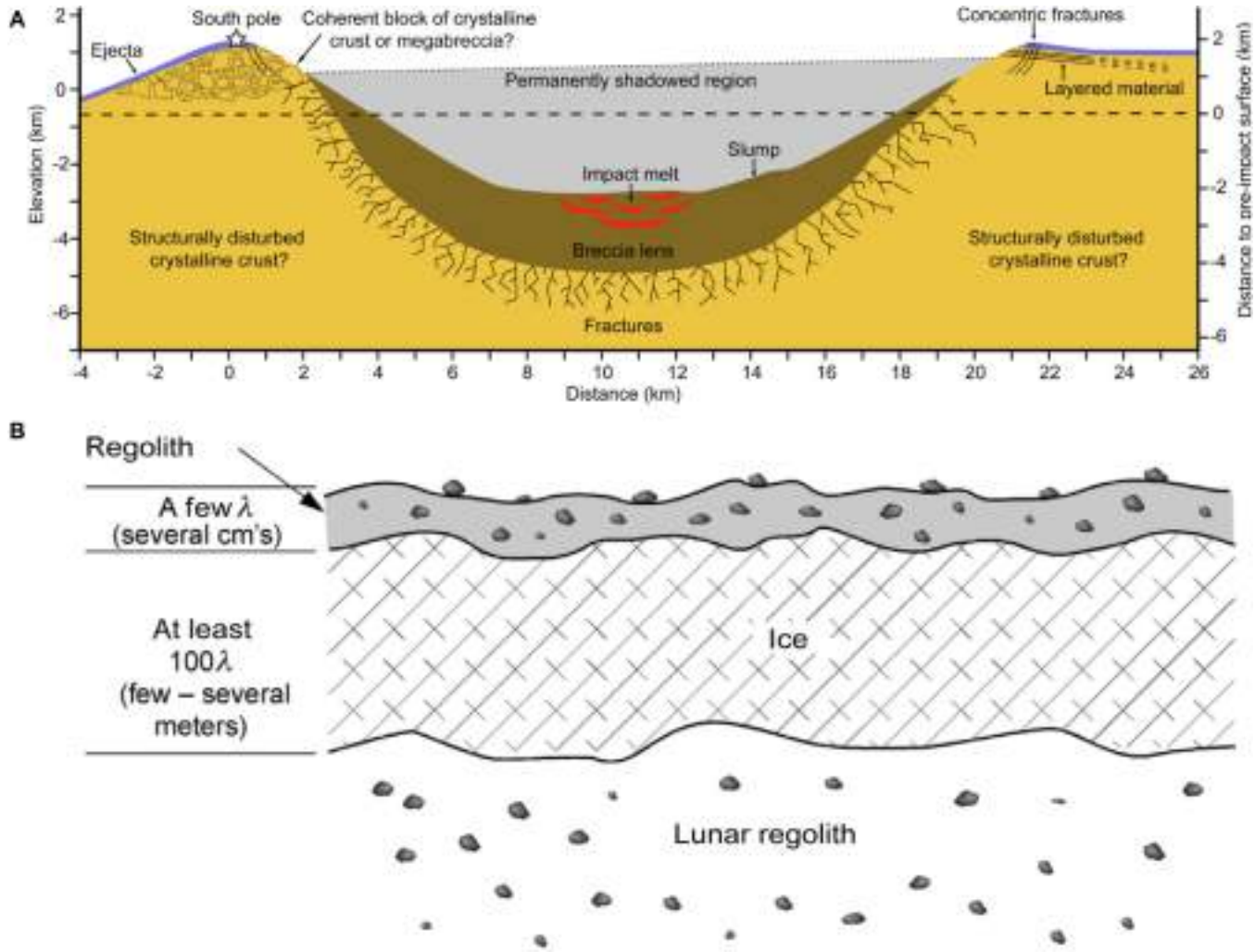

Fig. 7. (A) Cross-sectional view of the Shackleton crater [57]. (B) Presumed shallow surface structure of the lunar polar region containing water ice [21].

[5]. If the rover lands outside the crater with the lander, it will be difficult to enter the PSR by moving; if the landing site is chosen inside the Shackleton crater, it will face the problems of energy supply and measurement and control communication, and it will be difficult for the rover to leave the Shackleton crater after completing the exploration mission. We consider that the CE-7 leaper is a more ideal instrument, which can land in the PSR at the bottom of the Shackleton crater by flying, then fly to the lighted area outside the crater after completing the in situ exploration.

CE-7 mission detectors mainly include a lunar orbiter, a lander, a rover, and a leaper [4]. Both the orbiter and rover will carry a radar, respectively. The orbiter's radar will be a SAR system, while the rover's radar will likely inherit the GPR carried by Chang'E-3 and -4 rovers. However, its operating frequency has not yet been reported. If the presence of water ice in the PSR of the Shackleton crater is to be detected in the future, this paper also suggests that the leaper should also carry an in situ detection radar. When the leaper is working at the bottom of the Shackleton crater, the GPR can acquire in situ detection data from the shallow subsurface of the Shackleton crater. The observed data from GPR can be used to analyze the presence and reserves of water ice in the shallow subsurface layers.

Prospect of Water Ice in the Shackleton Crater Observed by the In Situ GPR

As mentioned in the "Evidence of Water Ice in the Shackleton Crater" section, there is a consensus that water ice is present in the PSR of the Shackleton crater by different detection methods. However, the form and distribution of water ice in the shallow surface layer and its dielectric properties need to be further investigated. Among the various detection methods, the in situ GPR equipment is expected to be used to detect the water ice in the shallow surface layer of the PSR in the Shackleton crater. The Chinese Chang'E-3 mission is the first one to use the GPR equipment for in situ observation of extraterrestrial objects [64,65]. Later, Chang'E-4 and -5, as well as Mars' Tianwen-1 and rover Perseverance, carried GPR equipment to penetrate the planet's surface and search for water ice [58,61,62,66–70].

The Moon-based GPRs onboard Chang'E-3 and -4 are nanosecond radars operating in the time domain with separate transmit and receive antennas, with the ultra-broadband nanosecond pulse generated by the transmit antenna of the radar, which propagates into the lunar subsurface material (Fig. 9). If the radar pulse encounters an inhomogeneous material or an interface with a different medium, it will be reflected and scattered, and then the echo signal will be received by the receiving

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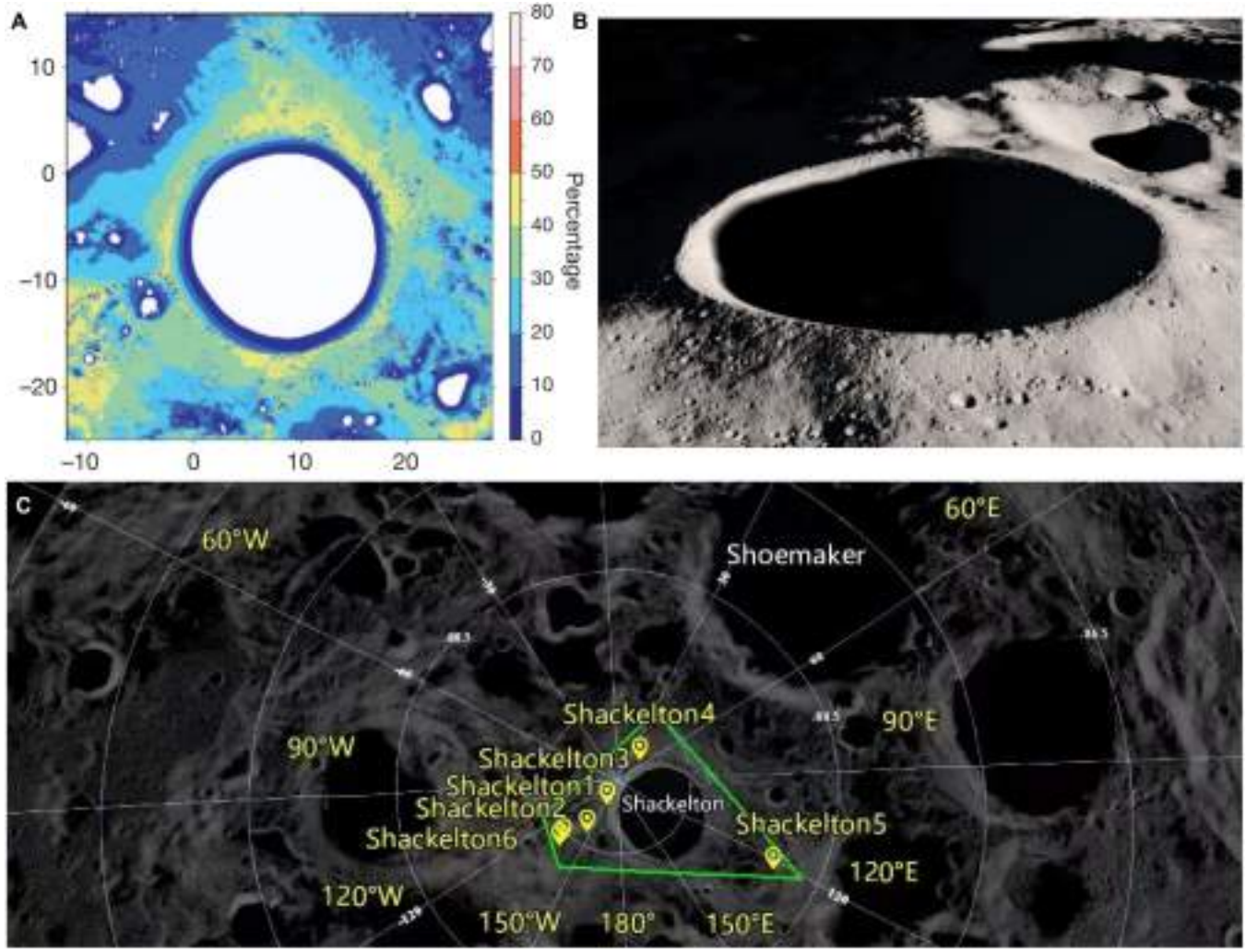

Fig. 8. Potential landing area for CE-7 mission: the south pole of the Moon. (A) Percentage of light time in the Shackleton crater [9]. (B) Image of the Shackleton crater. Source: NASA's Scientific Visualization Studio. (C) Image of the area around Shackleton crater. Source: Deep Space Exploration Laboratory of China in 2023.

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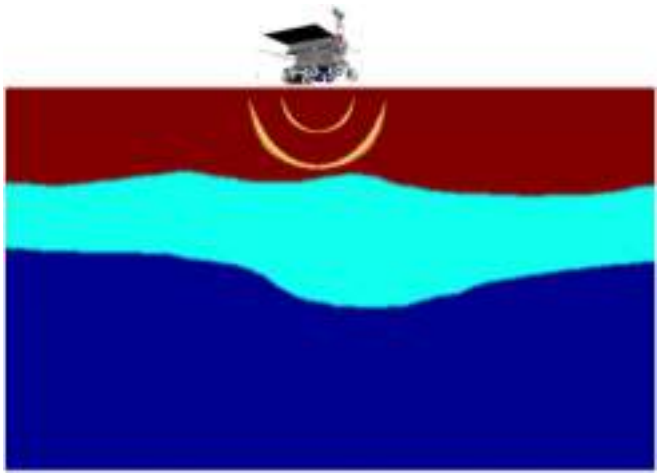

Fig. 9. A schematic diagram illustrating the assumed working mode of the GPR carried by the future CE-7 mission. A possible scenario for subsurface structures could be described as follows: The red medium denotes the lunar regolith covering, the green area represents water ice, and the blue area indicates the lunar regolith or ejecta.

antenna of the radar [27,64,65,71]. By processing, analyzing, and calculating the radar echo signals, it is possible to obtain the thickness of the lunar regolith, the dielectric properties of subsurface materials, and the geological structure of the subsurface layers along the lunar rover's path [27,62,64,72,73].

In general, the presence of water ice in the shallow surface layer can be determined based on GPR observation by inversing the dielectric properties of the subsurface material, i.e., the dielectric constant and the loss tangent. For pure water ice, the dielectric constant and loss tangent are generally ~3 and <0.01, respectively [74,75]. The Martian MARSIS radar has observed the internal medium in the Medusae Fossae Formation (MFF) region of the equator with a dielectric constant of ~2.9 and a loss angle tangent of ~0.0048 [76], which suggests that the MFF region of the equator may be rich in water ice, but it cannot be ruled out that the interior is composed of volcanic ash or low-density material [76,77]. The reason is that the water ice and the Martian soil's dielectric properties are very similar. Meanwhile, the dielectric properties of lunar regolith are also very similar to those of water ice; for example, the Chang'E-4 radar calculated a dielectric constant of ~3 and a loss tangent of ~0.005 for the

Table 1. Basic parameters of Chang'E-3, -4, and -5 in situ Moon-based GPRs [58,64,65,75,79]

Mission	Channel	Center frequency (MHz)	Frequency band (MHz)	Depth resolution (cm)	Penetrating depth (m)
CE-3/4	Channel 1	60	40–80	Meter scale	>100
	Channel 2	500	250–750	≤30	>30
CE-5		2,000	1,000–3,000	~5	~3

shallow surface fine-grained lunar regolith [62]. Therefore, it is difficult to distinguish the difference between fine-grained lunar regolith and water ice in terms of dielectric properties, which makes it a challenging task for GPR to observe water ice in the PSR at the bottom of the Shackleton crater.

How can we solve this problem? We suggest to consider the following aspects:

1. The first step is to calculate the dielectric properties of the shallow subsurface material based on the observed GPR data. For example, if the loss tangent of the material of the lunar subsurface is calculated to be less than 10^{-3}, it can be judged that water ice probably exists in the detection area. This is because water ice is a medium that is extremely lossy to radar electromagnetic waves. If the loss tangent of the subsurface material is >0.01, it indicates that the material in the detection area is likely to be dense basaltic material or lunar regolith with a high chemical composition of $TiO_2 + FeO$ [60], and the probability of the presence of water ice is not high.

2. The second step is to observe whether the echo features of the GPR radar image are clean or not. In general, the interior of lunar regolith is not an isotropic and inhomogeneous material [73]. It contains varying amounts of small rock fragments, which result in multiple reflections of radar electromagnetic waves [61,71,73]. In contrast, the interior of water ice is almost isotropic and homogeneous, and its radar echo features are cleaner.

3. GPR observation of the subsurface material of the Shackleton crater operating in multiple frequency channels. Experimental data show that the attenuation of electromagnetic waves in pure water ice is independent of frequency, while the attenuation in basalt sand or other different ratio mixtures is frequency dependent [78]. The analysis of the attenuation of radar electromagnetic waves in the detection region at different frequencies will be the key to proving the existence of water ice in the Shackleton crater using the in situ GPR observation.

Channel 1 of the Moon-based GPR onboard the Chang'E-3 mission has a center frequency of 60 MHz and a frequency band of 40 to 80 MHz, while Channel 2 has a center frequency of 500 MHz and a frequency band of 250 to 750 MHz [64]. As the backup of Chang'E-3, Chang'E-4 has the same GPR as Chang'E-3 [79]. The Chang'E-5 lander carries a lunar regolith penetrating radar with array antennas. Its operating frequency is 2,000 MHz, and the frequency band is 1,000 to 3,000 MHz [58,66,69]. The basic parameters of the previous Moon-based GPRs are listed in Table 1. It can be seen that the frequency of radar antennas carried by China's lunar exploration project missions is diverse. It has the ability to observe the shallow subsurface of the Moon using the GPR with different frequency channels. Therefore, we suggest that the GPR carried by the future CE-7 mission can be equipped with antennas operating at least 3 different frequencies for detecting water ice in the Shackleton crater.

Conclusion

The unique lighting conditions and surface environment in the polar regions of the Moon make them an ideal location for the accumulation and preservation of water ice and also an important target for future lunar exploration, especially the Chinese CE-7 mission [3,5,80]. This paper provides an overview of the research and exploration achievements in the Shackleton crater, located in the polar regions of the Moon. It presents the geological features, surface environmental characteristics, and the distribution of water ice within it [9]. The integrated results from different detection methods strongly suggest the presence of water ice in the PSR of the Shackleton crater. However, the distribution and existence state of water ice within the lunar regolith still lacks a unified and clear understanding at present. The GPR instrument to be carried out by the CE-7 mission will be proposed to detect the presence of water ice in the south pole PSR [4]. Understanding the distribution and existence states of water ice within the lunar regolith of the south pole through exploration is of significant engineering importance for assessing the total water reserves and for the extraction and utilization of water resources. Furthermore, the exploration of the polar regions will undoubtedly enhance our understanding of the geological activities of the Moon, and even the composition of early Solar System materials. The exploration and utilization of water and solar resources will also provide a solid foundation for establishing Moon bases and conducting more distant space explorations [5,81,82].

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Data Availability

This is a review paper; no new data were generated for this paper.

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